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For each major and year combination, calculate:
- Number of students
- Average exam score
- Average hours studied this is the question I am trying to solve

Your task is a **groupby on two columns (`major` and `year`) with multiple aggregations**. The clean and correct way to solve it in pandas is:

✅ Correct Solution

Python

Run

```python
result = df.groupby(['major', 'year']).agg(
    Num_Stud=('student_id', 'count'),
    Avg_exam_score=('exam_score', 'mean'),
    Avg_hours_studied=('hours_studied', 'mean')
)

print(result)
```

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#   - Number of students
#   - Average exam score
#   - Average hours studied
Num_Stud= df.groupby('major')['year'].agg(['student_id', 'count'])
Avg_exam_score=df.groupby('major') ['year'].agg(['exam_score', 'mean'])
Avg_hours_studied=df.groupby('major') ['year'].agg(['hours_studied', 'mean'])
print(Num_Stud)
print(Avg_exam_score)
print(Avg_hours_studied) why si this code giving me an error

Your code is giving an error because you are mixing up **columns and aggregation functions** in `.agg()`.

❌ The main problem

You wrote things like:

Python

```python
df.groupby('major')['year'].agg(['student_id', 'count'])
```

Issues:

1. You are grouping `year`, but trying to compute `student_id` → ❌ `student_id` is not a function
2. `agg()` expects **functions like** `"count"`, `"mean"` ...t column names like `"student_id"`
3. You are also applying `mean` to `"year"` instead of the correct columns (`exam_score`, `hours_studied`)

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✅ Correct understanding

You said you want:

- Number of students
- Average exam score
- Average hours studied

So your columns should be something like:

- `student_id`
- `exam_score`
- `hours_studied`
- `major`

✅ FIXED CODE

1. Number of students per major

Python

```python
Num_Stud = df.groupby('major')['student_id'].count()
```

---

2. Average exam score per major

Python

```python
Avg_exam_score = df.groupby('major')['exam_score'].mean()
```

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TODO: Create a 2x2 subplot figure containing:
1. Bar chart: Average exam score by major
2. Count plot: Number of students by year
3. Heat map: Correlation matrix of numerical columns
4. Violin plot: Exam score distribution by performance category

For this task, you'll typically use **Matplotlib**, **Seaborn**, and **pandas**.

Here's a complete example:

Python

Run

```python
import matplotlib.pyplot as plt
import seaborn as sns

# Create 2x2 subplot figure
fig, axes = plt.subplots(2, 2, figsize=(14, 10))

# 1. Bar chart: Average exam score by major
avg_scores = df.groupby('major')['exam_score'].mean().reset_index()

sns.barplot(
    data=avg_scores,
    x='major',
    y='exam_score',
    ax=axes[0, 0]
)

axes[0, 0].set_title('Average Exam Score by Major')
```

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Python

Run

```
# 3. Heatmap: Correlation matrix of numerical columns
numeric_cols = df.select_dtypes(include='number')

corr_matrix = numeric_cols.corr()

sns.heatmap(
    corr_matrix,
    annot=True,
    cmap='coolwarm',
    ax=axes[1, 0]
)

axes[1, 0].set_title('Correlation Matrix')

# 4. Violin plot: Exam score distribution by performance category
sns.violinplot(
    data=df,
    x='performance_category',
    y='exam_score',
    ax=axes[1, 1]
)

axes[1, 1].set_title('Exam Score Distribution by Performance Category')

# Adjust layout
plt.tight_layout()

# Show figure
```

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How do you calculate a correlation between two columns in a dataframe

To calculate **correlation between two columns in a Pandas DataFrame**, you use:

Python

```python
df['column1'].corr(df['column2'])
```

🧠 Example

Python

```python
df['hours_studied'].corr(df['exam_score'])
```

👉 This computes the **Pearson correlation coefficient** between the two columns.

📊 What the result means

The output is a number between **-1 and +1**:

Value	Meanir

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how do you create a new column in a dataframe grouping some categories and giving them same label

To create a **new column in a DataFrame that groups categories and assigns them the same label**, you typically use either:

- `map()` (simple)
- `np.where()` (conditional logic)
- `pd.cut()` (numeric ranges)
- `replace()` (manual mapping)

🧠 1. Best and simplest: `map()` (categorical grouping)

Example: grouping majors

Python

```python
df['major_group'] = df['major'].map({
    'CS': 'STEM',
    'Math': 'STEM',
    'Physics': 'STEM',
    'Biology': 'LIFE_SCIENCE'
})
```

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Create a new column 'performance' with categories:
'Excellent' (>90), 'Good' (80-90), 'Average' (70-80), 'Needs Improvement' (<70) I mean something like this

Got it — this is exactly a **range-based grouping problem**, so the cleanest way is `pd.cut()`.

✅ BEST SOLUTION (recommended)

Python

```python
df['performance'] = pd.cut(
    df['exam_score'],
    bins=[-np.inf, 70, 80, 90, np.inf],
    labels=['Needs Improvement', 'Average', 'Good', 'Excellent']
)
```

🧠 How it works

📊 Bins (ranges)

Range	Label
-∞ to 70	eds Improvement

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how do you create a pivot table

In Python, you create a pivot table using **pandas** with `pd.pivot_table()`.

---

1. Basic steps

Step 1: Import pandas

Python

```python
import pandas as pd
```

Step 2: Create or load your data

Python

```python
df = pd.DataFrame({
    "Region": ["East", "West", "East", "West"],
    "Product": ["Laptop", "Laptop", "Phone", "Phone"],
    "Sales": [1000, 1500, 800, 1200]
})
```

Step 3: Create the pivot table

Python

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I have only see the codes for lpotting one histogram or box plot but what I want to do right now I want to create a figure with 2 subplots so how do i go about it

Great question. When you want **multiple plots in the same figure**, you first create a figure and a set of axes (subplots), then tell each plot which axis to draw on.

📊 Create 2 subplots side by side

Python

```python
import matplotlib.pyplot as plt
import seaborn as sns

fig, axes = plt.subplots(1, 2, figsize=(12, 5))
```

What this means:

Python

```python
plt.subplots(1, 2)
```

creates:

- 1 row
- 2 columns

Visually:

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